



Cover illustration

Special issue: High-throughput process development.

High-throughput process development (HTPD) combines high-throughput experimentation, adequate analytical methods and advanced experimental design and/or data analysis. This Special issue is edited by Jürgen Hubbuch (Karlsruhe Institute of Technology, KIT, Karlsruhe, Germany). Cover image by Florian Dismer und Stefan Oelmeier showing a robot pipetting in 96-well format.

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Technical Report

Predicting protein dynamic binding capacity from batch adsorption tests

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The dynamic binding capacity (DBC) and its dependence on residence time influence the design and productivity of adsorption columns used in protein capture applications. This paper offers a very simple approach to predict the DBC of an adsorption column based on a measurement of the equilibrium binding capacity (EBC) and of the time needed to achieve one-half of the EBC in a batch adsorption test. The approach is based on a mass transfer kinetics model that assumes pore diffusion with a rectangular isotherm; however, the same approach is also shown to work for other systems where solute transport inside the particle occurs through other transport mechanisms.

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1 Introduction

Batch adsorption tests, increasingly conducted in high-throughput-screening (HTS) format [1], are used extensively to identify conditions that are suitable for chromatographic capture of proteins and other macromolecules. Such tests have the advantage of requiring small amounts of adsorbent and protein and can be conducted in parallel using either 96-well formats or mini-columns. When performed over sufficiently long times, batch tests easily provide information about the equilibrium binding capacity (EBC, also referred to as “static capacity”). Over shorter times, they also provide information on the kinetics of the adsorption process. Together, this information can be used to determine the residence time that in a column operation will provide a desired dynamic binding capacity (DBC). This is usually achieved through a model that describes the mass transfer mechanism [2, 3].

The objective of this paper is to provide explicit functions that describe the relationship between the time needed to attain one half of the EBC in a batch adsorption experiment and the residence time required to obtain a

specified DBC in a column. Exact expressions are provided for the case where the adsorption isotherm is rectangular and intraparticle mass transfer is controlled by pore diffusion. However, we show that, as long as intraparticle diffusion is controlling, the same relationship can be applied approximately for other cases so that no knowledge about transport mechanisms or particle size is actually needed. The approach is validated through a few examples based on published data.

2 Theory

2.1 Dynamic binding capacity

The DBC is defined as the amount of protein bound in a column when the outlet reaches a certain percentage of the feed concentration. The DBC is always a fraction of the equilibrium binding capacity (EBC), which is attained when all of the binding sites are occupied. For mass transfer controlled conditions, the ratio DBC/EBC depends only on the residence time (L/u) [3]. This ratio is also independent of feed concentration when the adsorption isotherm is rectangular and intraparticle mass transfer is controlled by pore diffusion. For this case, the following relationships are obtained [3] based on the general breakthrough expression of Weber and Chakraborti [4] valid for both constant pattern and non-constant pattern conditions:

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Abbreviations: DBC, dynamic binding capacity; EBC, equilibrium binding capacity

$$\frac{\text{DBC}|_{10\%}}{\text{EBC}} = \begin{cases} 0.364n - 0.0612n^2 + 0.00423n^3 & \text{for } n < 2.5 \\ 1 - 1.03/n & \text{for } n > 2.5 \end{cases} \quad (1)$$

$$\frac{\text{DBC}|_{1\%}}{\text{EBC}} = \begin{cases} 0.288n - 0.0307n^2 + 0.00083n^3 & \text{for } n < 2.5 \\ 1 - 1.19/n & \text{for } n > 2.5 \end{cases} \quad (2)$$

The first of these equations is for breakthrough at 10% of the feed concentration, while the second is for 1%. In both cases, n is the number of transfer units, which is given by:

$$n = 60(1 - \varepsilon) \frac{D_e L}{d_p^2 u} \quad (3)$$

where ε is the column extraparticle void fraction, D_e the effective pore diffusivity, d_p the particle diameter, L the column length, and u the superficial velocity. The residence time L/u is also equal to V_c/Q , where V_c is the column volume and Q is the volumetric flow rate. Explicit forms of Eqs. (1) and (2) giving n as a function of DBC/EBC can also be obtained for $n < 2.5$ using analytical expressions for the solution of a cubic equation [5]. Figure 1 gives a plot of Eqs. (1) and (2). The difference between the two curves is small because for these conditions, the initial portion of the breakthrough curve is very steep. As seen from this figure, 90% of the EBC is attained when $n \sim 10$, while 50% of the EBC is obtained when $n \sim 2$. The choice of the actual DBC/EBC and, hence, residence time depends on a balance of processing time and costs. For a given amount of feed, operating with a DBC/EBC close to 1 will minimize column volume, but the processing time will be long because of the long residence time required. On the other hand, operating at low values of DBC/EBC will require a larger column but the process will be faster. In general, an intermediate value of DBC/EBC will provide maximum productivity [3, 6].

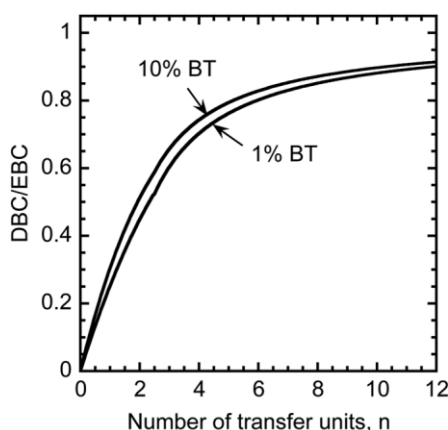


Figure 1. Relationship between DBC/EBC and the number of transfer units for a highly favorable isotherm. The two curves shown give DBC/EBC when the outlet concentration reaches 1 and 10% of the feed concentration, respectively. BT, breakthrough.

2.2 Batch adsorption kinetics

Batch adsorption tests are usually conducted by either measuring the residual protein concentration in solution at a given time and calculating the amount bound by material balance or by separating the particles from the solution, desorbing the protein in a known volume of desorbent, and determining the amount bound from the concentration in the desorbing solution. In either case, the amount of protein bound per unit of particle volume, $\bar{q}$, is obtained as a function of time for a given initial concentration C_0 . When the isotherm is rectangular and intra-particle mass transfer is controlled by pore diffusion, the relationship between $\bar{q}$ and time is given by the following equation [7]:

$$\frac{C_0}{q_m} \frac{D_e t}{d_p^2} = \frac{1}{4} I \quad (4)$$

where:

$$I = \frac{1}{6\lambda\Lambda} \ln \left[\frac{\lambda^3 + \eta^3}{\lambda^3 + 1} \left(\frac{\lambda + 1}{\lambda + \eta} \right)^3 \right] + \frac{1}{\lambda\Lambda\sqrt{3}} \left[\tan^{-1} \left(\frac{2\eta - \lambda}{\lambda\sqrt{3}} \right) - \tan^{-1} \left(\frac{2 - \lambda}{\lambda\sqrt{3}} \right) \right] - \frac{1}{3\Lambda} \ln \left(\frac{\lambda^3 + \eta^3}{\lambda^3 + 1} \right) \quad (4a)$$

$$\eta = \left(1 - \frac{\bar{q}}{q_m} \right)^{1/3} \quad (4b)$$

$$\lambda = \left(\frac{1}{\Lambda} - 1 \right)^{1/3} \quad (4c)$$

$$\Lambda = \frac{V_p q_m}{VC_0} \quad (4d)$$

V_p and V are the volumes of particles and solution, respectively, q_m is the binding capacity expressed per unit of particle volume, and C_0 is the initial protein concentration. Note that in order to ensure that complete saturation of the particles is possible, Λ must not be >1 . In practice, lower values of Λ are used when $\bar{q}$ is based on the residual protein solution concentration in order to minimize material balance errors that occur when the difference between initial and final protein concentration is small. For these conditions, the time required to achieve 50% of saturation of the adsorbent particles is:

$$t_{1/2} = \frac{I_{1/2}}{4} \frac{q_m}{C_0} \frac{d_p^2}{D_e} \quad (5)$$

where $I_{1/2}$ is the value of I obtained from Eq (4a) for $\bar{q}/q_m = 0.5$. This parameter depends only on Λ as is shown graphically in Fig. 2.

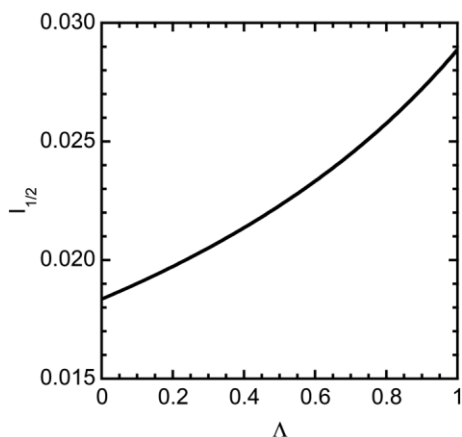


Figure 2. Plot of the function $I_{1/2}$ vs Λ given by Eq. (4d) for use in Eq. (5)–(8).

2.3 Relationship between batch kinetics and column performance

The relationship between the time in a batch experiment and the residence time needed to obtain a certain number of transfer units in a column and, hence, DBC, is obtained combining Eqs. (3) and (5), yielding:

$$\frac{L}{u} = \frac{1}{15I_{1/2}} \frac{n}{1-\epsilon} \frac{C_0}{q_m} t_{1/2} \quad (6)$$

Thus, experimentally measuring the time required to attain half-saturation of the adsorbent, $t_{1/2}$, and the binding capacity, q_m , allows one to calculate L/u . This calculation requires knowledge of the packed column extraparticle void fraction. In turn, this quantity can be measured experimentally or estimated from a suitable model for compressed packings [8–10]. The EBC is then given by $EBC = (1 - \epsilon)q_m$. Thus, Eq. (6) can also be written as follows:

$$\frac{L}{u} = \frac{n}{15I_{1/2}} \frac{C_0}{EBC} t_{1/2} \quad (7)$$

or

$$n = \frac{15I_{1/2}}{t_{1/2}} \frac{EBC}{C_0} \frac{L}{u} \quad (8)$$

It should be noted that these DBC expressions only consider the amount of protein actually held in the particles and neglect the amount of protein held in the voids outside the particles, while the actual DBC normally includes both. However, for practical conditions, with relatively dilute feeds, the amount held in the voids is usually a small fraction of the total.

3 Application examples

3.1 Example 1 – IgG capture by a protein A adsorbent

Figure 3A shows the batch uptake curve from [10] for the adsorption of 2 mg/mL IgG on the Protein A adsorbent UNOsphere SuPrA (Bio-Rad Laboratories, Hercules, CA, USA). For the experimental conditions of this batch adsorption test, $\Lambda = 0.55$, $q_m = 39$ mg/mL and $t_{1/2} = 0.93$ min. From Fig. 2, we have $I_{1/2} = 0.0228$. Taking $\epsilon = 0.35$ [10], Equation (7) yields values of $L/u = 2.1$ and 0.43 min for $n = 10$ and 2, respectively, which, as noted above, correspond to DBC values that are 90 and 50% of the EBC. The DBC predicted from Eq. (1) over the entire range of residence times is given by the solid line in Fig. 3B with the dashed lines showing the effects of a $\pm 10\%$ variation in the measured value of $t_{1/2}$. The DBC predictions are in good agreement with experimental data from [10] obtained for different feed concentrations and column lengths.

3.2 Example 2 – mAb capture by a macroporous cation exchanger

Figure 3C shows the batch uptake from [11] for the adsorption of 1 mg/mL of a mAb on the macroporous cation exchanger UNOsphere S (Bio-Rad Laboratories). For the experimental conditions of this batch adsorption test, $\Lambda = 0.51$, $q_m = 93$ mg/mL and $t_{1/2} = 4.4$ min. From Fig. 2, $I_{1/2} = 0.0224$. Taking $\epsilon = 0.30$ [12], Eq. (6) yields values of $L/u = 2.0$ and 0.40 min for $n = 10$ and 2, respectively. The DBC predicted from Eq. (1) over the entire range of residence times is given by the solid line in Fig. 3D with the dashed lines showing the effects of a $\pm 10\%$ variation in the measured value of $t_{1/2}$. The DBC predictions are in reasonable agreement with the experimental DBC data from [12] that are shown in this figure along with an estimated error of $\pm 10\%$.

3.3 Example 3 – mAb capture by a dextran-grafted cation exchanger

Figure 3E shows the batch uptake curve from [11] for the adsorption of 1 mg/mL of a mAb on the dextran-grafted cation exchanger Capto S (GE Healthcare, Piscataway, NJ, USA). For the experimental conditions of the batch adsorption test, $\Lambda = 0.45$, $q_m = 207$ mg/mL and $t_{1/2} = 3.5$ min. From Fig. 2, $I_{1/2} = 0.0228$. Taking $\epsilon = 0.30$ [12], Eq. (6) yields values of $L/u = 0.75$ and 0.15 min for $n = 10$ and 2, respectively. The DBC predicted from Eq. (1) over the entire range of residence times is given by the solid line in Fig. 3F with the dashed lines showing the effects of a $\pm 10\%$ variation in the measured value of $t_{1/2}$. The DBC predictions are in agreement with the experimental DBC data from [12] that are shown with an estimated er-

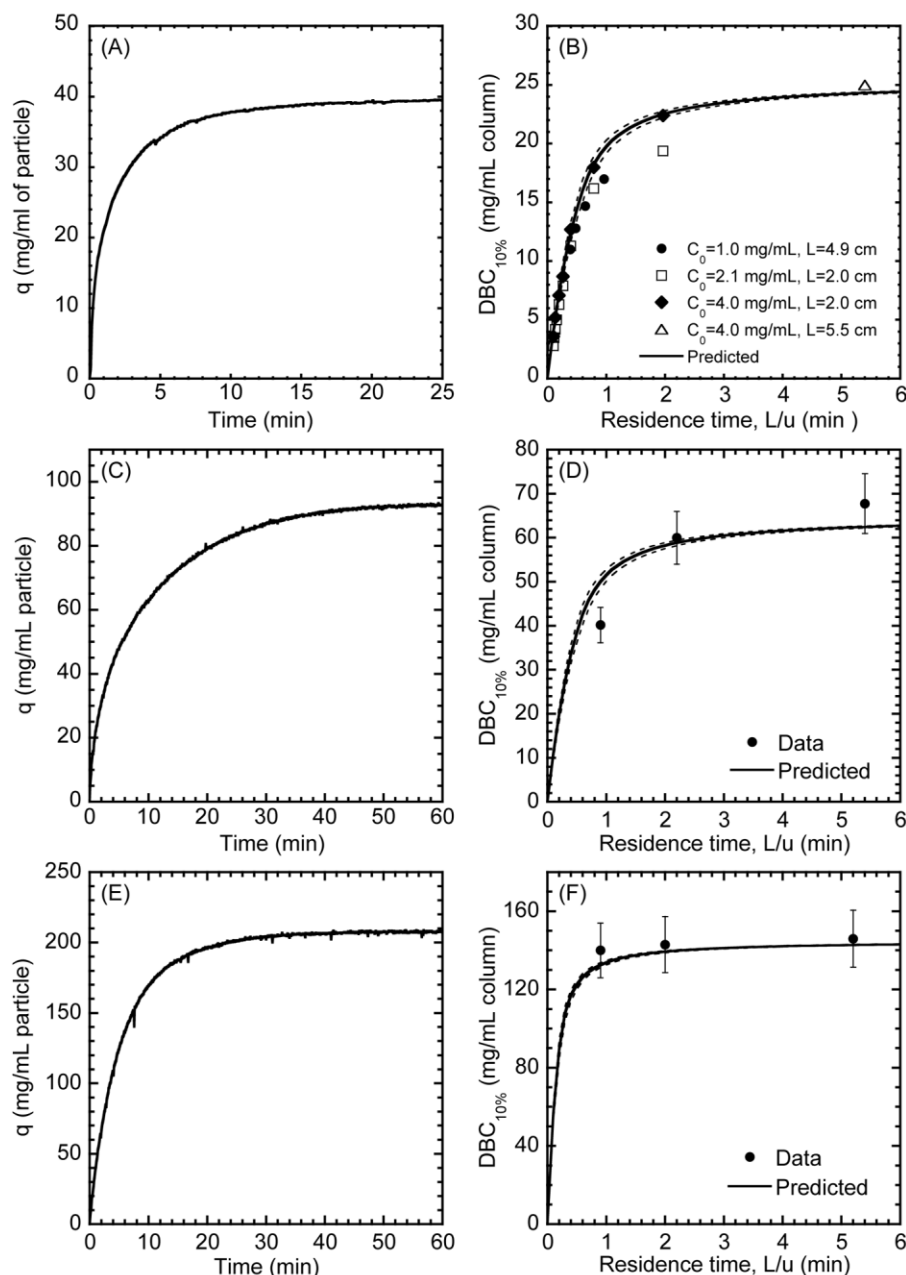


Figure 3. Experimental batch uptake and DBC_{10%} curves obtained for different systems compared to DBC predictions based on Equations 1 and 8. The dashed lines show predicted DBC values for a $\pm 10\%$ variation in $t_{1/2}$. (A) Batch uptake curve for 2 mg/mL IgG adsorption UNOsphere SuPrA and (B) corresponding DBC_{10%} data, both from [10]. (C) Batch uptake curve for 1 mg/mL mAb adsorption on UNOsphere S from [11] and (D) corresponding DBC_{10%} data from [12]. (E) Batch uptake curve for 1 mg/mL mAb adsorption on Capto S from [11] and (F) corresponding DBC_{10%} data from [12].

ror of $\pm 10\%$. Although the results predicted at very low residence times could not be verified experimentally for this column, since the velocity required would exceed the pressure limit, the predicted results are consistent with the experimental trend that near-equilibrium conditions (i.e., DBC~EBC) prevail even at residence times as low as 1 min. It should be noted that, for these conditions, protein mass transfer in Capto S is governed by a homogeneous or solid diffusion mechanism, which has a different diffusional driving force and is responsible for the fast uptake kinetics. Despite this difference, predictions based on Eq. (6) are in agreement with the data. This occurs because the breakthrough curve is quite insensitive to the

actual transport mechanism. Thus, the DBC predicted for different transport mechanisms is the same as long as n is the same [2, 3].

4 Conclusions

Simple equations based on a mass transfer model that assumes pore diffusion control with a rectangular isotherm predict the column DBC as a function of residence time based on a batch uptake test where the equilibrium capacity and the time needed to achieve one half of the equilibrium capacity are determined. The results are gen-

erally applicable regardless of the actual mass transfer mechanism, as long as mass transfer is controlling. Thus, given suitable experimental batch data, no knowledge of either mechanism or particle size is necessary for the practical purpose of translating batch times into the residence time that will give a desired DBC in a column.

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The author declares no conflict of interest.

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